

Plan: the round family — ellipse (O3-E) + capsule/stadium (O9-K)

HyMeKo manipulation program (Track A2, shapes)

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1 Scope and goal

Add the **round / smooth-non-circular** shape family — the complement of the polygon corner family: **O3-E**, a flat elliptical prism, and **O9-K**, a flat stadium (discorectangle) prism = the planar analogue of a capsule. Both are smooth-rimmed (no corners), orientation-dependent, and equal-area to the coin (mass = O0). O3-E fills the roadmap’s long-parked “O3 ellipse” slot.

Key recon result — native geoms are wrong here. MuJoCo’s native **ellipsoid** and **capsule** are 3-D rounded bodies that *bulge in z*; matched in the plane they still have a different *volume* than the flat coin cylinder, so they do **not** give mass parity (measured: ellipsoid 0.0335, capsule 0.0344 vs coin 0.0503 kg). They also break the flat-topped-puck invariant every existing shape shares. $\Rightarrow$ both round shapes are built as **flat mesh prisms** (a 2-D smooth boundary extruded to $\pm$ half-thickness), exactly like the polygon prisms — giving equal-area mass parity and consistent flat-face contact.

2 Geometry (verified numerically)

- **Ellipse** O3-E: boundary $(r_x \cos t, r_y \sin t)$, $N=128$ segments, extruded. Semi-axes $r_x=\text{radius}$, $r_y=\text{radius_y}$. Equal-area when $r_x r_y = r^2$. Chosen $r_x=0.025$, $r_y=0.016$ ($r_x r_y = 4 \times 10^{-4} = r^2$, aspect 1.56:1): mass 0.050246 ($\Delta_{O0}=2.0 \times 10^{-5}$), footprint 25.0 mm.
- **Capsule / stadium** O9-K: a rectangle with two semicircular caps of radius b ; half-length a , $\rho=a/b$. Area $4(a-b)b + \pi b^2 = \pi r^2 \Rightarrow b = r \sqrt{\pi / (4(\rho-1) + \pi)}$. For $\rho=2$: $a=0.026533$, $b=0.013266$; $n_{\text{cap}}=48$ arc segments/cap: mass 0.050250 ($\Delta_{O0}=1.6 \times 10^{-5}$), footprint 26.5 mm. Flat sides + round ends (distinct from the all-curved ellipse).

Both $\Delta_{O0} < 10^{-4}$ (the mass-parity tolerance) — clean shape-only ablations.

3 Affected files

- `hymeko_rl/env/object_spec.py`: `Shape.ELLIPSE`, `Shape.CAPSULE`; preconditions (both require `radius_y`; capsule requires `radius > radius_y`, i.e. $a > b$); footprint branches (ellipse $\rightarrow \max(r_x, r_y)$, capsule $\rightarrow a=\text{radius}$). **No new field** — both reuse the existing `radius/radius_y` (already threaded via `compose_kwargs`).
- `hymeko_rl/env/planar_grasp_env.py`: `_ellipse_prism_vertices`, `_stadium_prism_vertices`; `compose_planar_scene` branches “ellipse” / “capsule” (reuse `disk_radius_y`; **no new kwarg**).
- **NEW** `data/robotics/galampos_env_o3_ellipse.hymeko`, `..._o9_capsule.hymeko`.

- `hymeko_rl/coin_delivery/object_curriculum.py`: add O3-E, O9-K.
- tests (`test_object_spec.py`, `test_object_curriculum.py`); this plan; report dir.

4 CORE.YAML items touched

None. `object_spec.py/planar_grasp_env.py` are non-core; `compose_planar_scene` is the sanctioned builder (no new caller). No dependency added.

5 Interface changes

Additive: two new **Shape** members + two new geom branches. **No signature change** (ellipse/capsule reuse `radius/radius_y`). Existing shapes' code paths are untouched (new `elif` branches only) — so the regression risk is lower than the N-gon refactor.

6 Test strategy

- Unit (`object_spec`): `from_str` ellipse/capsule; preconditions (missing `radius_y` asserts; capsule $a \leq b$ asserts); `polygon_sides()` is `None`; footprint values.
- Unit (`object_curriculum`): membership (10); O3-E/O9-K geom **MESH**, mass $\approx$ O0, footprint.
- Integration: generation smoke over the full 10-variant curriculum — 0 model/contract failures, **O0–O8 certified-capture counts unchanged** vs the N-gon run (the new branches must not perturb existing shapes), O3-E/O9-K generate + certify straddle.
- Static: `ruff`, `mypy --strict` on `object_spec`; ownership-guard green.
- Performance: peak RSS < 2 GB; record wall.

7 Performance budget & risk

Peak RSS < 2 GB (cap 16 GB). Wall: 10-variant smoke $\approx$ 40 min CPU (60 rollouts). Risks: (i) mass parity — the mesh is an N -segment approximation; mitigated by $N=128/n_{\text{cap}}=48$ (measured $\Delta < 2 \times 10^{-5}$) and asserted $< 10^{-4}$. (ii) Existing-shape regression — low (additive branches); the smoke's O0–O8 count match is the gate; halt if any drifts. (iii) Capsule degeneracy ($a \leq b$) rejected by precondition. (iv) Worst-case input: full 10-variant smoke, no training.

8 Rollback

Single commit; revert drops the two **Shape** members, the two generators + branches, the two scenes, the two curriculum entries. No state mutation.

9 Out of scope

No teacher / retrieval / RL / delivery — generation + capture characterization only. Higher-eccentricity ellipses / other stadia are trivial follow-ups (just new scene literals).